# Nidhi Kumari

AI & Machine Learning Engineer | Aspiring AI Architect
+91-6287205761 | nidhitiwaryy011@gmail.com | LinkedIn | github.com/nidhi713

## Professional Summary

Computer Science engineer (B.Tech, CGPA 7.88/10) with a clear career goal: to grow into an **AI Architect** who designs intelligent systems end to end – from data pipelines and models to agents and real-time deployment. Hands-on across Deep Learning, Computer Vision, and LLM-powered multi-agent workflows (LangGraph, Gemini API), backed by an active AI internship on **Megha Mitra** at BMSK Patna, a Grade-A internship at CDAC Patna, and 200+ solved DSA problems. I pair strong fundamentals with a systems mindset, building AI that moves from raw data to a working, user-facing solution.

## Technical Skills

**Languages:** Python, C++, SQL
**AI/ML:** Machine Learning, Deep Learning, Computer Vision, CNN, LSTM, Transfer Learning, Attention / Multiple Instance Learning
**Generative AI:** LLMs, Prompt Engineering, AI Agents, Multi-Agent Workflows (LangGraph), Gemini API
**Libraries:** PyTorch, TensorFlow/Keras, OpenCV, MediaPipe, Scikit-learn, Pandas, NumPy, Matplotlib
**Tools:** Git, GitHub, Jupyter Notebook, VS Code, MySQL
**Core CS:** DSA, OOPs, DBMS
**Soft Skills:** Problem Solving, Communication, Collaboration

## Experience

**AI Intern** – *Megha Mitra Project* 2026 – Present
*BMSK Patna*

- Contributing to **Megha Mitra**, an AI project at BMSK Patna, working on AI/ML model development and its integration into the application.
- Applying Python, machine learning and Generative AI techniques to build, test and refine intelligent features in a team setting.

**AI & Data Science Intern** Dec 2025 – Jan 2026
*CDAC Patna* (Grade A)

- Developed ML models in Python covering preprocessing, feature engineering, training and evaluation.
- Built AI workflows for data cleaning, visualization and predictive modeling using Pandas, NumPy, Scikit-learn and GitHub.

## Projects

**AI Data Analyst Agent** *Python, LangGraph, Gemini API*

- Built a multi-agent AI system (LangGraph) that analyzes datasets, generates insights and answers questions in natural language.
- Automated analytics workflows using Pandas, Matplotlib and the Gemini API.

**Real-Time Sign Language Detection** *Python, TensorFlow, MediaPipe, OpenCV*

- Built a real-time gesture recognition system that translates sign language from live webcam video into text, as an assistive tool for the hearing impaired.
- Extracted 1,662 pose, face and hand keypoints per frame with MediaPipe Holistic; trained a 3-layer stacked LSTM on 30-frame sequences to classify gestures.
- Created a custom dataset and a live OpenCV inference pipeline with on-screen prediction confidence; easily extensible to new signs.

**Vision-Based Mind Wandering Detection** *PyTorch, ResNet18, CNN*

- Developed a deep learning pipeline to detect mind wandering from facial video sequences.
- Extracted frame-level features with pretrained ResNet18 and Attention-based Multiple Instance Learning (Video, Frames, Face Crop, ResNet18, Attention Pooling, Classification).

## Education

**B.Tech in Computer Science Engineering** 2022 – 2026
Government Engineering College, Kaimur, Bihar – **CGPA: 7.88/10**

## Certifications & Achievements

- AI & Data Science Internship (Grade A) – CDAC Patna; Data Analysis Certification – ExcelR.
- Solved 200+ Data Structures and Algorithms problems.
- Built AI projects spanning Machine Learning, Deep Learning, Computer Vision and AI Agents; actively exploring LLM applications and automation.